

SOURAV RAJ

Patna, Bihar

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PROFESSIONAL SUMMARY

Aspiring **Data Scientist** with hands-on experience in **Machine Learning, Deep Learning, NLP, LLM Applications, Agentic AI, and MLOps**. Experienced in building production-grade AI applications, predictive models, and end-to-end ML pipelines using **Python, SQL, Pytorch, FastAPI, LangGraph, Docker, AWS, and Google Cloud**.

PROJECTS

AI Blog Generation Platform (LLMs & Agentic AI)

Apr 2026 – May 2026

- Developed a production-ready **AI Blog Generation Platform** using **FastAPI, LangGraph, and LangChain**, implementing a **multi-agent workflow** for web research, blog planning, long-form content generation, AI image generation, and publishing.
- Designed a modular AI content generation pipeline supporting **OpenAI, Google Gemini, and Groq**, enabling backend-configurable LLM selection with **Tavily Search**-based web research.
- Built scalable backend services with **Async SQLAlchemy, PostgreSQL (Neon), Alembic, and JWT authentication**, implementing secure user management and asynchronous database operations.
- Automated cloud deployment using **Docker, GitHub Actions, Google Cloud Run, Artifact Registry, Secret Manager, and OIDC**, enabling secure CI/CD with database migrations.

Real Estate Price Prediction System (ML & Dashboard)

Oct 2025 – Dec 2025

- Web-scraped and aggregated **40K+ real estate listings** from **MagicBricks** (for educational purposes); performed extensive **data cleaning, missing value imputation, and feature engineering** to prepare a high-quality modeling dataset.
- Engineered **13 predictive features** improving model robustness and trained a **LightGBM regression model** achieving **$R^2 = 0.90$ and $MAE \approx 0.6$ Cr INR** using Optuna-driven hyperparameter tuning.
- Built an end-to-end **ML pipeline** using **MLflow, DVC, Docker, and AWS (S3, EC2, ECR)** with **GitHub Actions CI/CD** for automated experiment tracking, versioning, and deployment.
- Deployed a **Streamlit web application** enabling real-time property price prediction and interactive dashboards for market insights.

Twitter Sentiment Analysis (NLP & Deep Learning)

Aug 2025 – Present

- Performed EDA on **69K+ tweets** to analyze class imbalance, text distribution, and noise patterns for model design decisions.
- Developed a **PyTorch-based LSTM model** achieving **88% accuracy, 88% macro F1-score, and 0.98 AUC** on the test dataset.
- Integrated **MLflow with Optuna** for experiment tracking and tuning.
- Deployed inference application on **Streamlit Cloud** enabling real-time sentiment predictions.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, SciPy), SQL (MySQL)

Machine Learning: Scikit-learn, XGBoost, LightGBM, Regression, Classification, Clustering, Model Evaluation

Deep Learning: PyTorch, TensorFlow, Keras

Statistics: Hypothesis Testing, Feature Engineering, A/B Testing

MLOps & Deployment: MLflow, DVC, Docker, FastAPI

Cloud: AWS (EC2, S3, ECR, CodeDeploy)

Visualization: Matplotlib, Seaborn, Power BI, Excel

Version Control: Git, GitHub, GitHub Actions

Generative AI: LangChain, LangGraph, Retrieval-Augmented Generation (RAG)

EDUCATION

Bachelor of Science in Mathematics: Scored (76%)

2020 – 2023

Patna Science College, Patna

CERTIFICATIONS

Data Science Mentorship Program 2.0 – CampusX

Jan 2026

Machine Learning A-Z: AI, Python & R – Udemy

July 2025